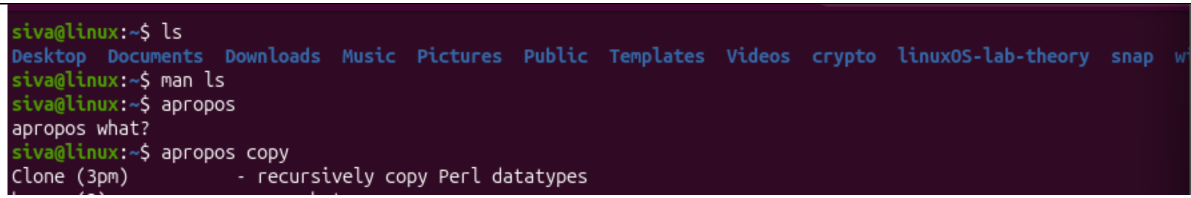
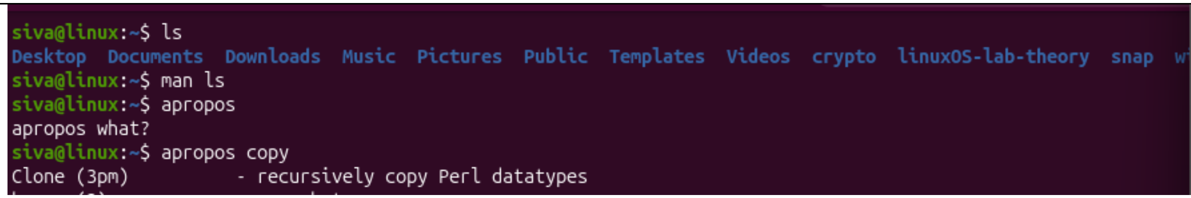
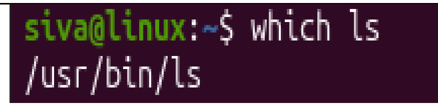
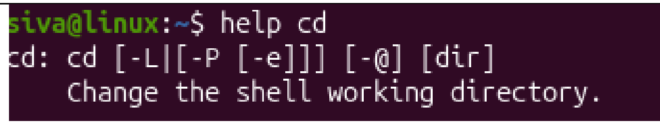
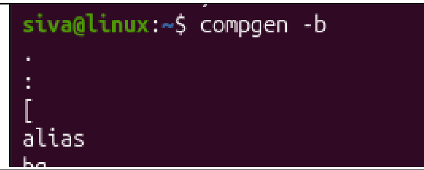
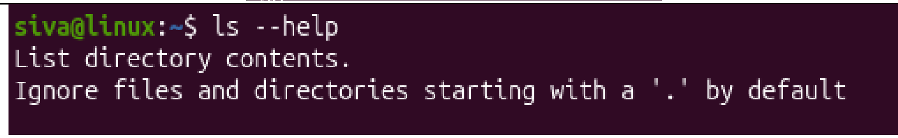
- **NAME: SIVARAMAKRISHNAN R**
- **ROLL NO: 261100690026**

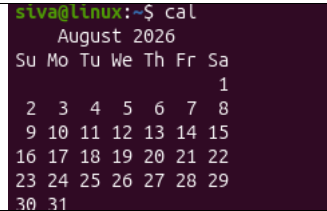
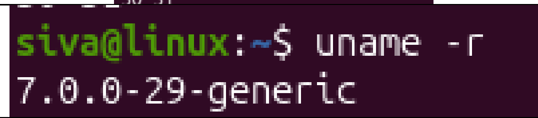
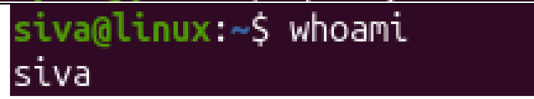
LINUX LAB ASSINGMENT 2

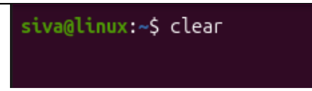
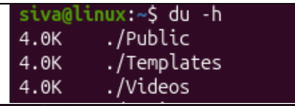
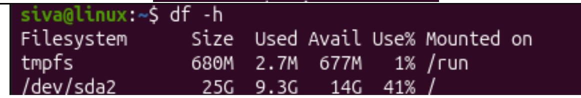
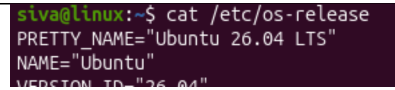
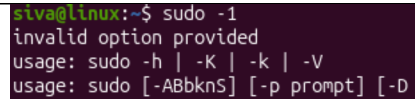
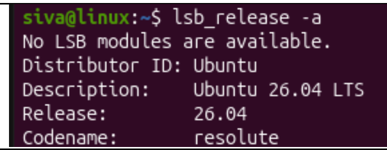
1. Getting Help

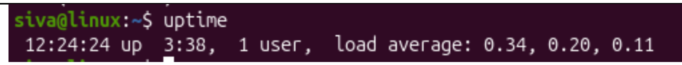
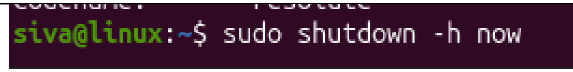
Task	Command Name	Syntax	Example	Screenshots
To get manual page for the known command	man ls	man <command>	man pwd	
To get manual page for the unknown command	apropos copy	apropos <keyword>	apropos copy	
To know the source file binary	whereis ls	whereis <command>	whereis ls	

To know the path of the command	which ls	which <command>	which ls	
To know the command is external or internal	type ls	type <command>	type cd	
To get help for the internal command	help cd	help <command>	help cd	
To list out bash commands	compgen -b	compgen -b	compgen -b	
To know the usage of the command	ls --help	<command> --help	ls --help	

2. Basic Commands

Task	Command Name	Syntax	Example	Screenshots
To know today's date	date	date	date	
To print calendar	cal	cal	cal	
To print kernel version	uname -r	uname -r	uname -r	
To print default shell	`echo $SHELL`	`echo $SHELL`	`echo $SHELL`	
To print currently logged in user	whoami	whoami	whoami	
To create shortcut for command	alias	alias <name>='<cmd>'	alias ll='ls -l'	
To delete shortcut	unalias	unalias <name>	unalias ll	

To change the timestamp of the file	touch	touch <file>	touch abc.txt	
To clear the screen	clear	clear	clear	
To create empty files	touch	Touch filename	touch 1.txt 2.txt	
To know disk usage	du	du -h	du -h	
To know free space in the system	df	df -h	df -h	
To know about the Linux release	cat	cat <file>	cat /etc/os-release	
To show login history	last	last	last	
To check the current user privileges	sudo	Sudo -l	Sudo -l	
To find about the Linux release	Lsb_release	Lsb_release -a	Lab_release	

To check system uptime	uptime	uptime	uptime	
To shutdown	shutdown	Sudo shutdown -h now	Sudo shutdown -h now	
To restart	reboot	Sudo reboot	Sudo reboot	

3. Navigation

Task	Command	Syntax	Example	Screenshot
To navigate home directory	cd	Cd ~	Cd ~	
To navigate to the parent directory	cd	Cd ..	Cd ..	

To navigate to the child directory	cd	Cd <directory>	Cd file1	siva@linux:~/Downloads$ cd pract1
Alternate command to cd	pushd	pushd <directory>	pushd file1	siva@linux:~/Downloads$ pushd pract1 ~/Downloads/pract1 ~/Downloads
To go back to the previous directory	Cd -	Cd -	Cd -	siva@linux:~/Downloads/pract1$ cd - /home/siva/Downloads
To go to the root directory	Cd	Cd /	Cd /	siva@linux:~/Downloads$ cd / siva@linux:/$

4. **File System**

Task	Syntax	Command	Screenshot
How to identify the file system	**df -T [path]**	**df -T .**	siva@linux:/$ df -T . Filesystem Type 1K-blocks Used Available Use% Mounted on /dev/sda2 ext4 25623780 9723292 14573540 41% /

5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

Use **globbing patterns wherever possible**, rather than specifying filenames individually.

a. List all files ending in .html.

b. List all files **other than .html files** using extended globbing.

c. List filenames containing at least one numeric digit.

d. List files whose names begin with user followed by a single digit.

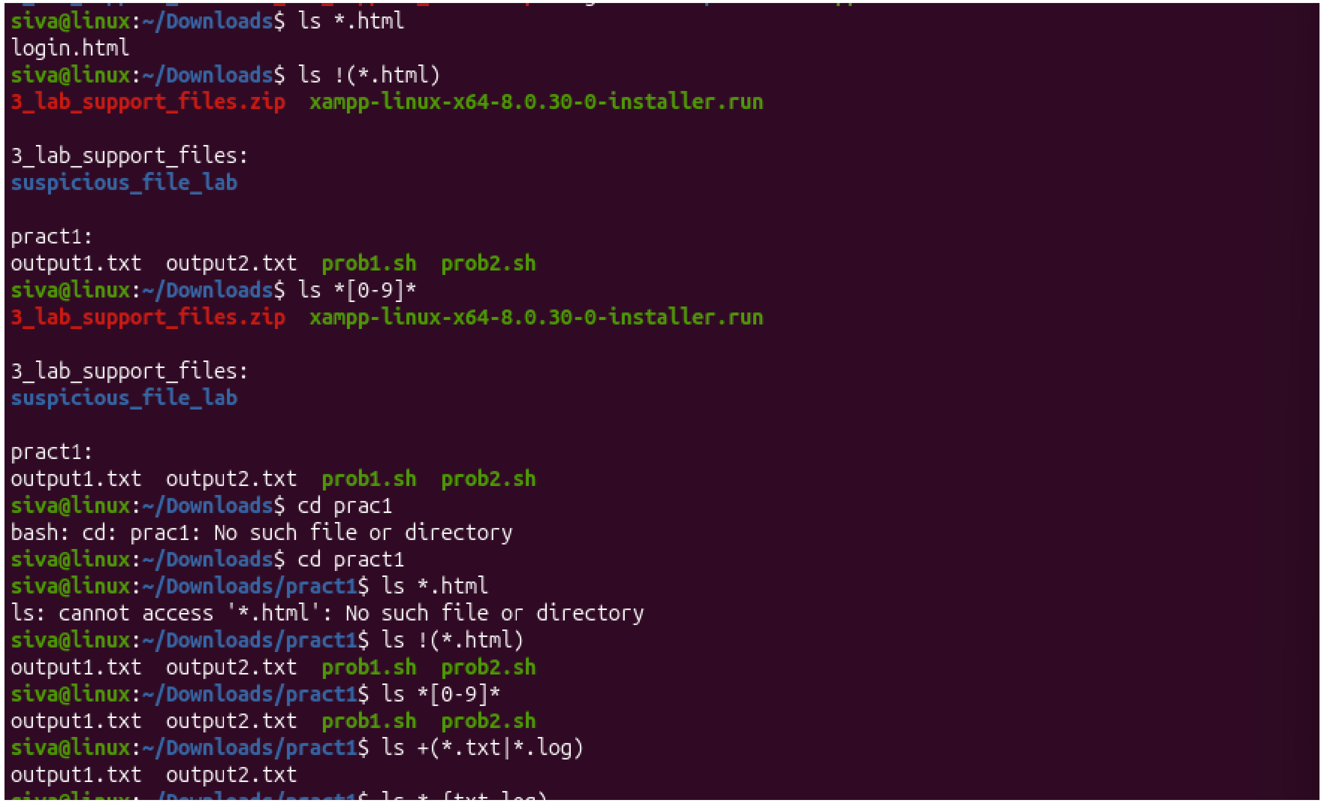
e. List files ending in either .txt or .log using a single **extended glob pattern**.

f. List files that do **not** end in .txt or .log.

g. List files beginning with either access or error using extended globbing.

h. List .html files whose filenames contain a numeric digit.

NO	Method	Command
1	List all files ending in .html.	ls *.html
2	List all files other than .html files using extended globbing.	ls !(*.html)
3	List filenames containing at least one numeric digit.	ls *[0-9]*
4	List files whose names begin with user followed by a singledigit.	ls user[0-9].*
5	List files ending in either .txt or .log using a single extended glob pattern.	ls +(*.txt|*.log)
6	List files that do not end in .txt or .log	ls *.{txt,log}
7	List files beginning with either access or error using extended globbing.	ls {access*,error*}
8	List .html files whose filenames contain a numeric digit.	ls *[0-9]*.html



```
siva@linux:~/Downloads$ ls *.html
login.html
siva@linux:~/Downloads$ ls !(*.html)
3_lab_support_files.zip  xampp-linux-x64-8.0.30-0-installer.run

3_lab_support_files:
suspicious_file_lab

pract1:
output1.txt  output2.txt  prob1.sh  prob2.sh
siva@linux:~/Downloads$ ls *[0-9]*
3_lab_support_files.zip  xampp-linux-x64-8.0.30-0-installer.run

3_lab_support_files:
suspicious_file_lab

pract1:
output1.txt  output2.txt  prob1.sh  prob2.sh
siva@linux:~/Downloads$ cd prac1
bash: cd: prac1: No such file or directory
siva@linux:~/Downloads$ cd pract1
siva@linux:~/Downloads/pract1$ ls *.html
ls: cannot access '*.html': No such file or directory
siva@linux:~/Downloads/pract1$ ls !(*.html)
output1.txt  output2.txt  prob1.sh  prob2.sh
siva@linux:~/Downloads/pract1$ ls *[0-9]*
output1.txt  output2.txt  prob1.sh  prob2.sh
siva@linux:~/Downloads/pract1$ ls +(*.txt|*.log)
output1.txt  output2.txt
```

6. Extended Globbing

a. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

Write an extended glob pattern that selects only access.log, error.log, and auth.log.

Select all files except notes.txt.

Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.

No	Method	Command
1	Write an extended glob pattern that selects only access.log, error.log, and auth.log.	ls @(access.log|error.log|auth.log)
2	Select all files except notes.txt.	ls !(@(notes.txt))
3	Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.	ls auth.log.@(1|2)

```
siva@linux:~/Downloads$ ls @(access|error|auth).log
access.log
siva@linux:~/Downloads$ ls !(notes.txt)
```

b. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log

construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.

```
siva@linux:~/Downloads$ touch malware1.log malware2.log malwareA.log malware_report.txt normal.log
siva@linux:~/Downloads$ ls malware*[0-9]*.log
malware1.log  malware2.log
siva@linux:~/Downloads$ 
```

No	Method	Command
1	construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.	ls malware*[0-9]*.log

c. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log

use extended globbing to select .log files containing either failed_login or successful_login.

Construct a pattern that lists all .log files **except** firewall.log.

No	Method	Command
1	use extended globbing to select .log files containing either failed_login or successful_login.	ls @(failed_login|successful_login).log
2	Construct a pattern that lists all .log files except firewall.log.	ls !(@(firewall.log))

```
siva@linux:~/Downloads$ touch successful_login.log
siva@linux:~/Downloads$ ls @(failed_login|successful_login).log
failed_login.log  successful_login.log
siva@linux:~/Downloads$ ls !(@(firewall.log))
3_lab_support_files.zip  error.log.auth.log.1  malware1.log        normal.log            sucessful_login.
access.log               failed.log            malware2.log        notes.txt             system.log
auth.log.2               failed_login.log      malwareA.log        successful_login.log  xampp-linux-x64-
auth1.log.1              login.html            malware_report.txt  sucessful.log
```

7. Given

No.	Task	Method	Command
1	Display first **10 entries** of login.log	head → first 10 lines	head -10 login.log
2	Display first **5 entries** of access.log	head → first 5 lines	head -5 access.log
3	Display first **15 security events** of security.log	head → first 15 lines	head -15 security.log
4	Display first **20 events** of incident.log	head → first 20 lines	head -20 incident.log
5	Display only **first entry** of login.log	head → first 1 line	head -1 login.log
6	Display first **25 web requests** of access.log	head → first 25 lines	head -25 access.log

7	Display last **10 entries** of login.log	tail → last 10 lines	tail -10 login.log
8	Display last **5 entries** of access.log	tail → last 5 lines	tail -5 access.log
9	Display **20 most recent** security events	tail → last 20 lines	tail -20 security.log
10	Display **25 most recent** incident events	tail → last 25 lines	tail -25 incident.log
11	Display only **last entry** of security.log	tail → last 1 line	tail -1 security.log
12	Display last **50 events** of incident.log	tail → last 50 lines	tail -50 incident.log
13	Display login.log from **line 41 to end**	tail → start from line 41	tail -n +41 login.log
14	Display security.log from **line 101 to end**	tail → start from line 101	tail -n +101 security.log
15	Find total **lines** in login.log	wc -l → count lines	wc -l login.log
16	Find total **lines** in access.log	wc -l → count lines	wc -l access.log

17	Find total **security events** in security.log	wc -l → count lines	wc -l security.log
18	Find total **events** in incident.log	wc -l → count lines	wc -l incident.log
19	Count **words** in login.log	wc -w → count words	wc -w login.log
20	Count **bytes** in security.log	wc -c → count bytes	wc -c security.log
21	Display **line, word, byte counts** of access.log	wc → all counts	wc access.log
22	Line counts of **all .log files**	wc -l + wildcard	wc -l *.log
23	First **20** login.log → last **5**	head → tail	head -20 login.log | tail -5
24	Last **20** login.log → first **5**	tail → head	tail -20 login.log | head -5
25	First **30** access.log → last **10**	head → tail	head -30 access.log | tail -10
26	Last **30** access.log → first **10**	tail → head	tail -30 access.log | head -10
27	First **50** security events → last **10**	head → tail	head -50 security.log | tail -10

28	Last **50** security events → first **10**	tail → head	tail -50 security.log | head -10
29	Display lines **11–20** of login.log	First 20 → last 10	head -20 login.log | tail -10
30	Display lines **21–30** of login.log	First 30 → last 10	head -30 login.log | tail -10
31	Display lines **11–20** of access.log	First 20 → last 10	head -20 access.log | tail -10
32	Display lines **31–40** of access.log	First 40 → last 10	head -40 access.log | tail -10
33	Display security events **21–30**	First 30 → last 10	head -30 security.log | tail -10
34	Display security events **51–60**	First 60 → last 10	head -60 security.log | tail -10
35	Display incident events **101–110**	First 110 → last 10	head -110 incident.log | tail -10
36	Display incident events **201–220**	First 220 → last 20	head -220 incident.log | tail -20

37	Display incident events **491–500**	First 500 → last 10	head -500 incident.log | tail -10
38	First **20** login.log → count lines	head → wc -l	head -20 login.log | wc -l
39	Last **15** access.log → count lines	tail → wc -l	tail -15 access.log | wc -l
40	First **50** security events → count words	head → wc -w	head -50 security.log | wc -w
41	Last **25** security events → count lines	tail → wc -l	tail -25 security.log | wc -l
42	Extract lines **21–30** of login.log → count	head → tail → wc -l	head -30 login.log | tail -10 | wc -l
43	Extract lines **51–60** of access.log → count	head → tail → wc -l	head -60 access.log | tail -10 | wc -l
44	Extract events **101–150** → count	head → tail → wc -l	head -150 incident.log | tail -50 | wc -l
45	First 100 → last 20 → count lines	head → tail → wc -l	head -100 incident.log | tail -20 | wc -l